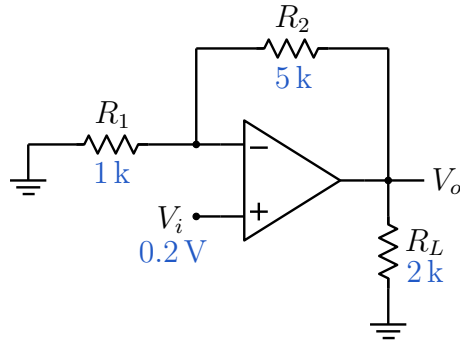


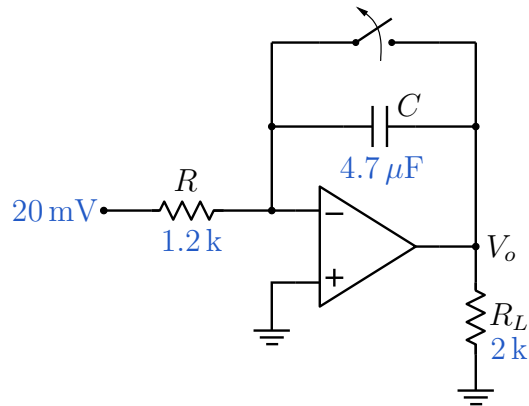
Basic Electronics

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Assignment for Week 7

1. In the circuit shown in the figure, the resistors have a tolerance of $\pm 1\%$. If the op-amp is ideal, what is the approximate range of V_o ?

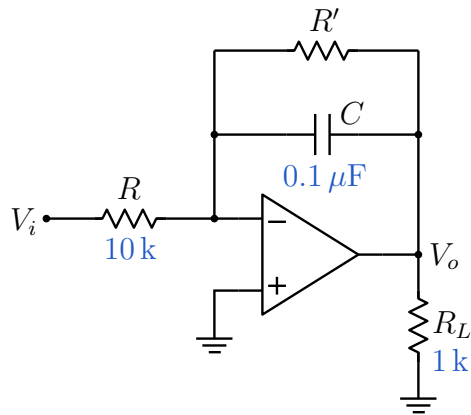


- (A) 1.1 V to 1.3 V (B) 1.15 V to 1.25 V (C) 1.18 V to 1.22 V (D) 1.19 V to 1.21 V
2. In the circuit shown in the figure, the op-amp saturation voltage is $\pm V_{\text{sat}} = \pm 12\text{ V}$. If the switch is opened at $t = 0$, at what time will the op-amp reach saturation?

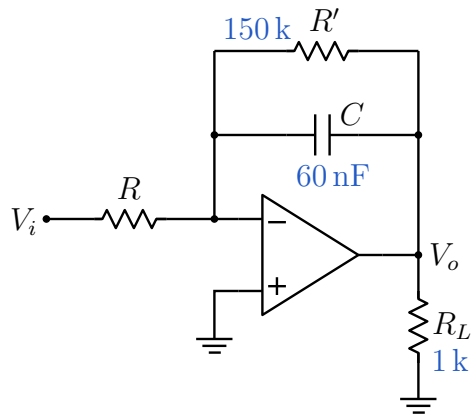


- (A) 3.4 sec (B) 0.58 sec (C) 63 msec (D) 9.2 msec

3. In the integrator circuit shown in the figure, we want to ensure that the impedance presented by the capacitor at a frequency of 5 kHz is smaller than R' by a factor of at least 50. What is $R'_{\min}$?

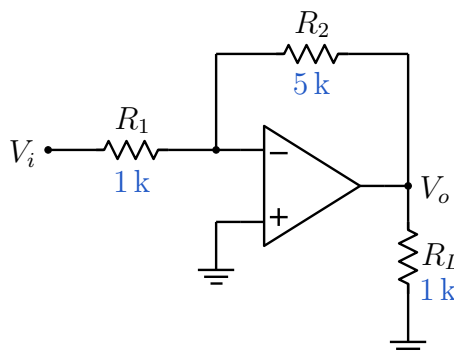


- (A) 2.5 kΩ (B) 106 kΩ (C) 68 kΩ (D) 15.9 kΩ
4. In the integrator circuit shown in the figure, the input is a 5 V peak-to-peak, 2 kHz square wave. What value of R is required for the output voltage to be 2 V peak-to-peak?

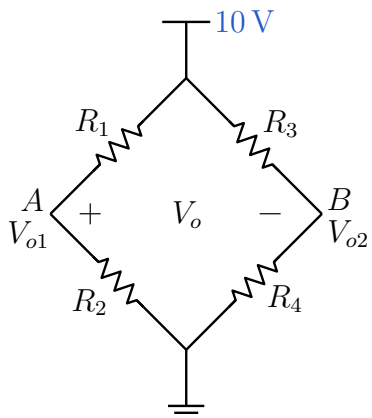


- (A) 5.2 kΩ (B) 20.8 kΩ (C) 7.1 kΩ (D) 12.8 kΩ

5. In the circuit shown in the figure, the op-amp offset voltage is in the range $-1\text{ mV} < V_{OS} < +1\text{ mV}$. If the op-amp is otherwise ideal, what is the range of the output voltage with $V_i = 20\text{ mV}$?

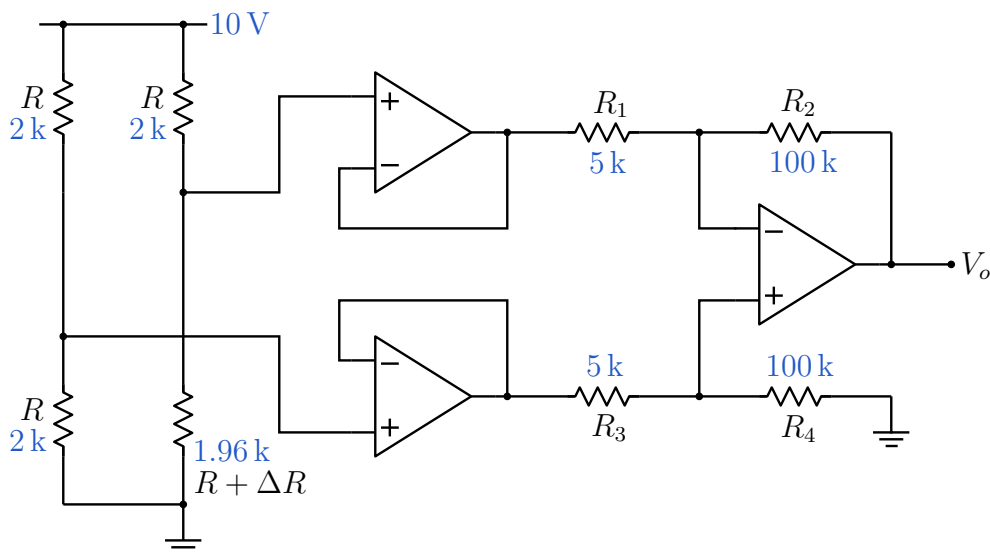


- (A) -112 mV to -88 mV
 (B) -106 mV to -94 mV
 (C) -110 mV to -90 mV
 (D) -102 mV to -98 mV
6. In the bridge circuit shown in the figure, $R_1 = R_2 = R_3 = R$, and R_4 lies in the range $0.92 R < R_4 < 1.08 R$. What is the common-mode voltage V_{oc} ?



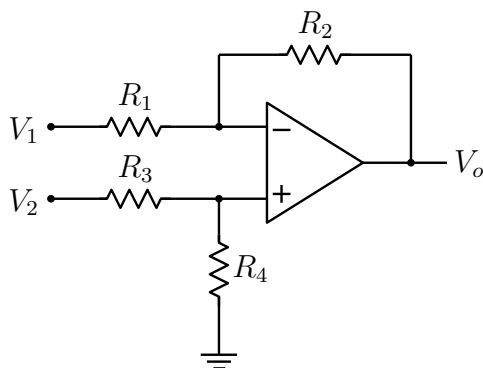
- (A) 5 V (B) 2.5 V (C) 7.5 V (D) 0 V
7. In the circuit of Q-6, what is the maximum value of $|V_{od}|$?
- (A) 0.1 V (B) 0.4 V (C) 0.2 V (D) 0.15 V

8. In the circuit shown in the figure, let V_{id} be the differential-mode input voltage for the difference amplifier, V_{ic} be the common-mode input voltage, V_{od} be the differential-mode output voltage, and V_{oc} be the common-mode output voltage. The nominal values of R_1 , R_2 , R_3 , R_4 are shown in the figure. What is $|V_{id}|$?



- (A) 12.5 mV (B) 25 mV (C) 37.5 mV (D) 50 mV
9. In the circuit of Q-8, suppose R_1 , R_2 , R_3 , R_4 are equal to their nominal values. What is V_{od} ?
- (A) 1 V (B) 0.5 V (C) 2 V (D) 1.25 V
10. In the circuit of Q-8, suppose $R_1 = 5 k\Omega$, $R_2 = R_4 = 100 k\Omega$, but R_3 is slightly different than its nominal value, viz., $R_3 = 5.04 k\Omega$. What is $|V_{oc}|$?
- (A) 50 mV (B) 38 mV (C) 68 mV (D) 92 mV
11. For the conditions given in Q-10, the CMRR of the difference amplifier is
- (A) 3,750 (B) 1,500 (C) 2,625 (D) 1,800
12. In the circuit of Q-8, suppose each of the resistances R_1 , R_2 , R_3 , R_4 , has a tolerance of 1%. What is the worst-case value of V_{oc} ?
- (A) 20 mV (B) 50 mV (C) 100 mV (D) 200 mV
13. For the conditions of Q-12, what is the worst-case CMRR of the difference amplifier?
- (A) 500 (B) 1,500 (C) 1,750 (D) 2,250

14. In the difference amplifier shown in the figure, $R_1 = R_3 = 10\text{ k}\Omega$, $R_2 = 100\text{ k}\Omega$. The resistance R_4 , which should be equal to R_2 , is slightly different, viz., $100.8\text{ k}\Omega$. What is the common-mode gain of the circuit?



- (A) 5.6×10^{-3} (B) 1.5×10^{-2} (C) 7.3×10^{-3} (D) 4.5×10^{-2}
15. For the difference amplifier of Q-14, what is the CMRR?
- (A) 2,600 (B) 1,375 (C) 980 (D) 1,750